

Ousama Jamal Eddin

Software Engineer | Flutter & Mobile Development

Jeddah, Saudi Arabia • +966 59 274 5955 • ossama.jamal.aldien@gmail.com

[GitHub](#) • [LinkedIn](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Computer Engineering graduate and software engineer with hands-on experience across mobile development, embedded systems, and full-stack projects. Specialized in building and deploying production-ready cross-platform mobile applications with Flutter, Dart, and Firebase. Owned the mobile app for a team-built, blockchain-based academic credential wallet (Ethereum), and shipped a published app to Google Play. Skilled in Riverpod state management, REST/Dio networking, Firebase Authentication, Cloud Firestore, and real-time data sync, with a strong foundation in clean architecture, data structures, and full app lifecycle delivery. Comfortable working independently in remote, asynchronous teams across multiple time zones.

TECHNICAL SKILLS

Languages: Dart, Python, Java, C++, C, Verilog, PHP, HTML/CSS, SQL

Mobile & Frameworks: Flutter (mobile & web), Dart, Riverpod (state management), Dio, Android, Material Design, Laravel (PHP, Blade), Bootstrap, HTML/CSS

Firebase & Backend: Firebase Authentication, Cloud Firestore, Cloud Functions, Realtime Database, REST APIs, Socket.IO, Node.js/Express, PostgreSQL, MySQL

Web3 & Cryptography: Ethereum, Solidity (integration), WalletConnect, EIP-712 signing, ECIES, decentralized identity & verifiable credentials

Concepts: State Management, Clean Architecture, Responsive UI/UX, Localization (i18n), Real-Time Systems, Debugging, App Deployment

Tools & Platforms: Git, GitHub, Google Play Console, Android Studio, VS Code, Vivado, MATLAB, Unity

EXPERIENCE

Engineering Intern at AvionChip Teknoloji

Jul 2024 – Aug 2024

Istanbul, Turkey

- Designed and implemented an RPN (Reverse Polish Notation) calculator in Verilog using the Xilinx Vivado toolchain on FPGA hardware.
- Built and optimized stack-based operations and core arithmetic functions (addition, multiplication), improving processing efficiency for accurate calculation execution.
- Handled serial data input/output and validated digital-circuit behavior through simulation and on-board testing.

Web Developer Intern, SCRAMBLEBIT

Jul 2025 – Aug 2025

Istanbul, Turkey

- Contributed to the design and development of dynamic web applications using Laravel and PHP, implementing Blade templates for efficient UI rendering.
- Structured backend functionality using Laravel's migrations, models, controllers, and routing, with MySQL for database management.
- Built responsive, user-friendly interfaces with HTML, CSS, and Bootstrap, and carried out debugging and testing to maintain code quality and optimize performance.

FEATURED PROJECTS

S2dent, Graduation Project (Blockchain-Based Academic Credential Wallet)

2025 – 2026

Decentralized system for issuing, holding, and verifying tamper-proof academic credentials on Ethereum. Owned the cross-platform Flutter mobile app (for students and verifiers) end-to-end within a 3-person team; shipped as v1.0.1.

- Architected and built the full Flutter/Dart mobile app using a feature-first clean architecture (data / domain / presentation) across 7 modules: auth, profile, credentials, QR verification, home, settings, and role selection.
- Implemented state management with Riverpod and a Dio-based REST networking layer, including a mock-API interceptor that let the app run fully without the backend to speed up development.

- Engineered on-chain identity and credential features: EIP-712 typed-data signing, ECIES decryption of encrypted credential payloads (PointyCastle), and WalletConnect wallet integration via Reown AppKit, with secure key/token storage using flutter_secure_storage.
- Built time-limited verification QR generation (qr_flutter) for students and a live camera scanner (mobile_scanner) for verifiers, returning real-time valid/invalid results checked against Ethereum smart contracts.
- Added real-time notifications over Socket.IO and an OTP-based authentication flow, with polished UI via FlexColorScheme theming, Google Fonts, and Lottie/SVG assets.
- Integrated the app within a 4-part monorepo (Flutter app, Next.js dashboard, Node.js/Express + PostgreSQL API, Solidity/Hardhat smart contracts), working against a role-based contract hierarchy with revocable, hash-unique credentials.

Technologies: Flutter, Dart, Riverpod, Dio, Reown AppKit (WalletConnect), EIP-712, ECIES, PointyCastle, flutter_secure_storage, mobile_scanner, qr_flutter, Socket.IO; integrated with Solidity / Ethereum (Hardhat) and a Node.js/Express + PostgreSQL backend.

Buzzkill (Flutter, in closed testing on Google Play)

A focus and productivity app built with Flutter, aimed at students who struggle with distraction while studying. Built around the Pomodoro technique with two modes: a fully customizable Solo Focus mode (session length, break count, and break duration), and an online Battle mode that matches two users into a shared 25-minute focus session, where forfeiting counts as a loss and finishing earns XP that levels up the user's profile and unlocks titles and icons.

- Designed and built both the solo and real-time multiplayer focus modes end-to-end in Flutter, including session timers, matchmaking, and win/loss/forfeit logic.
- Built an XP and leveling system with unlockable titles and icons to drive engagement and repeat use.
- Managed the full release workflow: build configuration, app signing, and tester distribution, currently in closed testing on the Google Play Console.

ADDITIONAL PROJECTS

Flutter & Firebase apps: Chat App (real-time messaging with Firebase Authentication and Cloud Firestore), Expense Tracker, Meals App, Favorite Places, and Shopping List, covering state management, navigation, local/cloud data, and responsive layouts.

CPU Implementation in Verilog (Team): Collaborated on designing a fully functional CPU, implementing the PC, IR, MAR, MDR, Control Unit, Accumulator, and ALU using hardware description language and digital design principles.

Portfolio website: Personal portfolio built with Flutter Web and deployed on Vercel (ousama-jd.vercel.app).

Game development: Flappy Bird clone (Unity, C#), plus Treasure Hunt and Hangman (Python).

EDUCATION

Fenerbahçe University, B.Sc. Computer Engineering
Istanbul, Turkey

2022 – 2026

Al-Thagher High School, High School Diploma
Jeddah, Saudi Arabia

2018 – 2020

CERTIFICATIONS & COURSES

- Flutter & Dart, The Complete Guide [2025 Edition]
- Foundations of Cybersecurity
- The Bits and Bytes of Computer Networking
- Technical Support Fundamentals
- MATLAB Onramp (MathWorks Training Services)

LANGUAGES

Arabic (Native) • English (Professional Proficiency) • Turkish (B1, Intermediate)